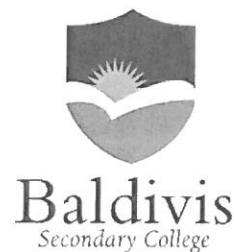


Year 9 Mini-Test 3
Probability and Linear Relationships

Resources allowed
Calculator and a page of Notes



Name: A. N. Sers Date: _____ Score: _____/40

1). If a 6 sided dice is rolled, the chance of getting a 5 is? [1 Mark]

- a) $\frac{1}{6}$ b) $\frac{5}{6}$ c) $\frac{6}{5}$ d) 1:6

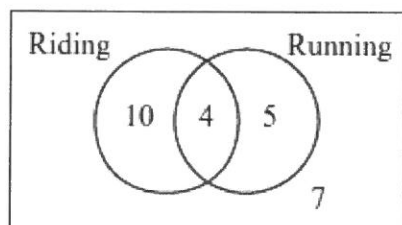
2). If a 6 sided dice is rolled, what is the chance of getting less than 4? [1 Mark]

- a) $\frac{1}{6}$ b) $\frac{1}{2}$ c) $\frac{2}{3}$ d) 4:6

3). If a 6 sided dice is rolled, what is the chance of getting the complement of rolling a 4? [1 Mark]

- b) $\frac{1}{6}$ b) $\frac{5}{6}$ c) $\frac{2}{3}$ d) 4:6

Use the following Venn diagram to complete the next 4 questions



4). How many people like running? [1 Mark]

- a) 7 b) 9 c) 10 d) 14 e) 4 f) 26

5). How many people like riding but don't like running? [1 Mark]

- a) 7 b) 9 c) 10 d) 14 e) 4 f) 26

6). How many like both running and riding? [1 Mark]

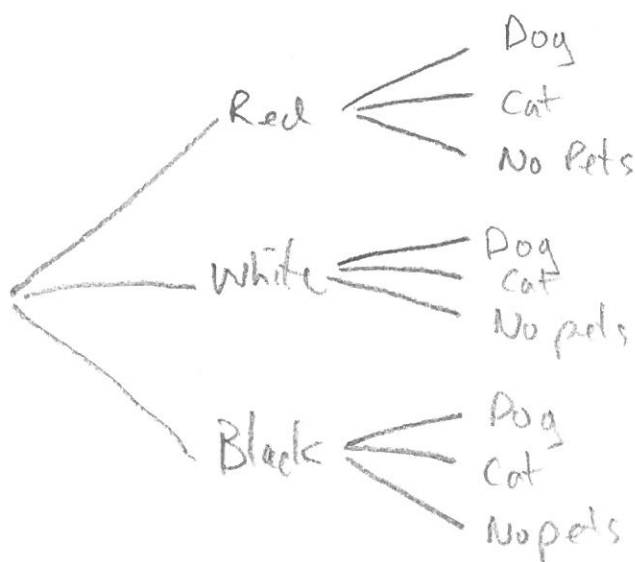
- a) 7 b) 9 c) 10 d) 14 e) 4 f) 26

7). How many people in total were asked if they like running or riding? [1 Mark]

- a) 7 b) 9 c) 10 d) 14 e) 4 f) 26

8). A family has an even chance of owning a red car, a white car or a black car. At the same time each family has an even chance of owning a dog, a cat or no pets.

a. Draw a tree diagram. [4 marks]



1 for correct
setting out

$\frac{1}{2}$ for errors.

b. What is the sample space of possible outcomes? [1 Mark]

$\{R+D, R+C, R+N, W+D, W+C, W+N, B+D, B+C, B+N\}$ ✓

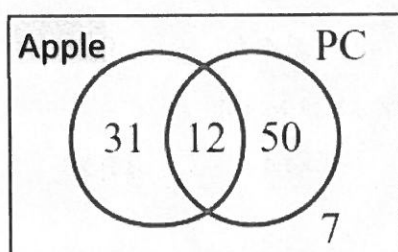
c. What is the probability the family owns a white car and a dog? [1 Mark]

$$\text{Prob}(W+D) = \frac{1}{9}$$
 ✓

d. What is the probability the family has a pet and a black car? [1 Mark]

$$\text{Prob}(B+C \text{ or } B+D) = \frac{2}{9}$$
 ✓

9. The Venn diagram below shows how many people own a PC computer, an Apple or both. Use this to complete the 2 way table. [3 Marks]

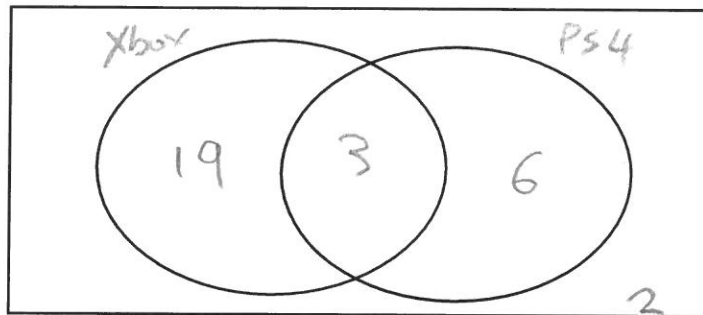


	Apple	No Apple	Total
PC	12	50	62
No PC	31	7	38
Total	43	57	100

$\frac{1}{2}$ for errors.

10. In a class of 30 students, 22 have an Xbox and 9 have a PS4 console. Three had both an Xbox and a PS4 console and 2 students carried neither.

a) Represent this using a Venn diagram. [3 marks]



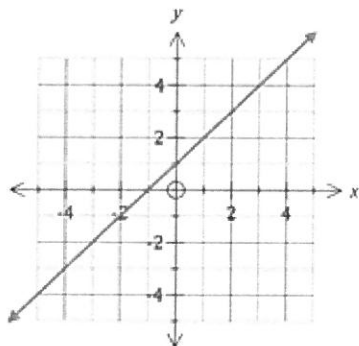
$-\frac{1}{2}$ for missing labels or wrong numbers

b) What is the probability that a student has both an Xbox and a PS4? [2 Mark]

$$\text{Prob(both)} = \frac{3}{30} = \frac{1}{10}$$

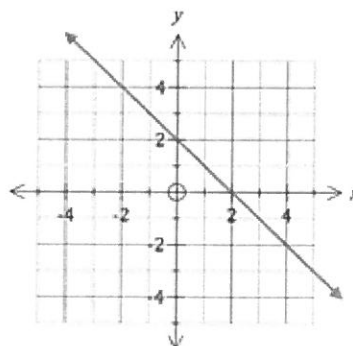
11. Find the gradient for each of the following. [1 Mark each = 3 marks]

A)

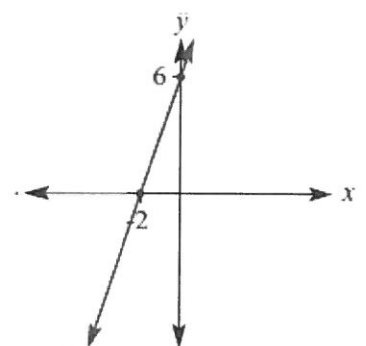


Gradient = 1

B)



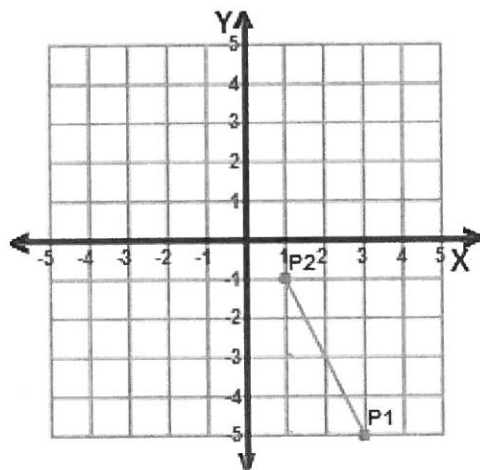
Gradient = -1



Gradient = 3

12. Find the length of the line and the coordinates for the mid point of the line for both of the following graphs [6 marks]

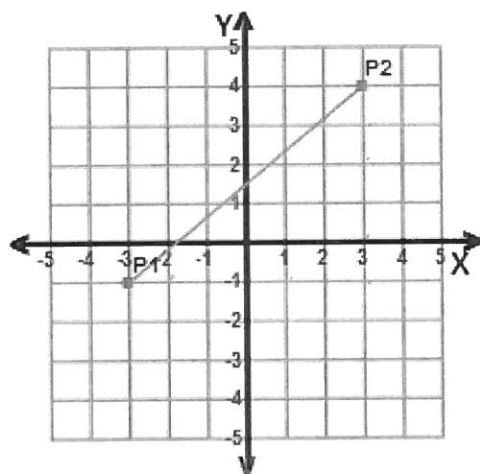
a)



$$\begin{aligned} \text{Length of the line} &= \sqrt{\text{rise}^2 + \text{run}^2} \\ &= \sqrt{4^2 + 2^2} \quad \checkmark \\ &= \sqrt{20} \text{ or } 4.47 \quad \checkmark \end{aligned}$$

Midpoint =

$$(2, -3) \quad \checkmark$$



$$\begin{aligned} \text{Length of the line} &= \sqrt{6^2 + 5^2} \quad \checkmark \\ &= \sqrt{61} \text{ or } 7.81 \quad \checkmark \end{aligned}$$

Midpoint =

$$(0, 1.5) \quad \checkmark$$

13. A survey of 25 000 new car buyers found that 750 had a major mechanical problem in the first year of operation. Calculate the relative frequency of:

a) having mechanical problems in the first year [1 mark]

$$rf = \frac{750}{25\,000} = 0.03 \quad \checkmark \text{ or } 3\%$$

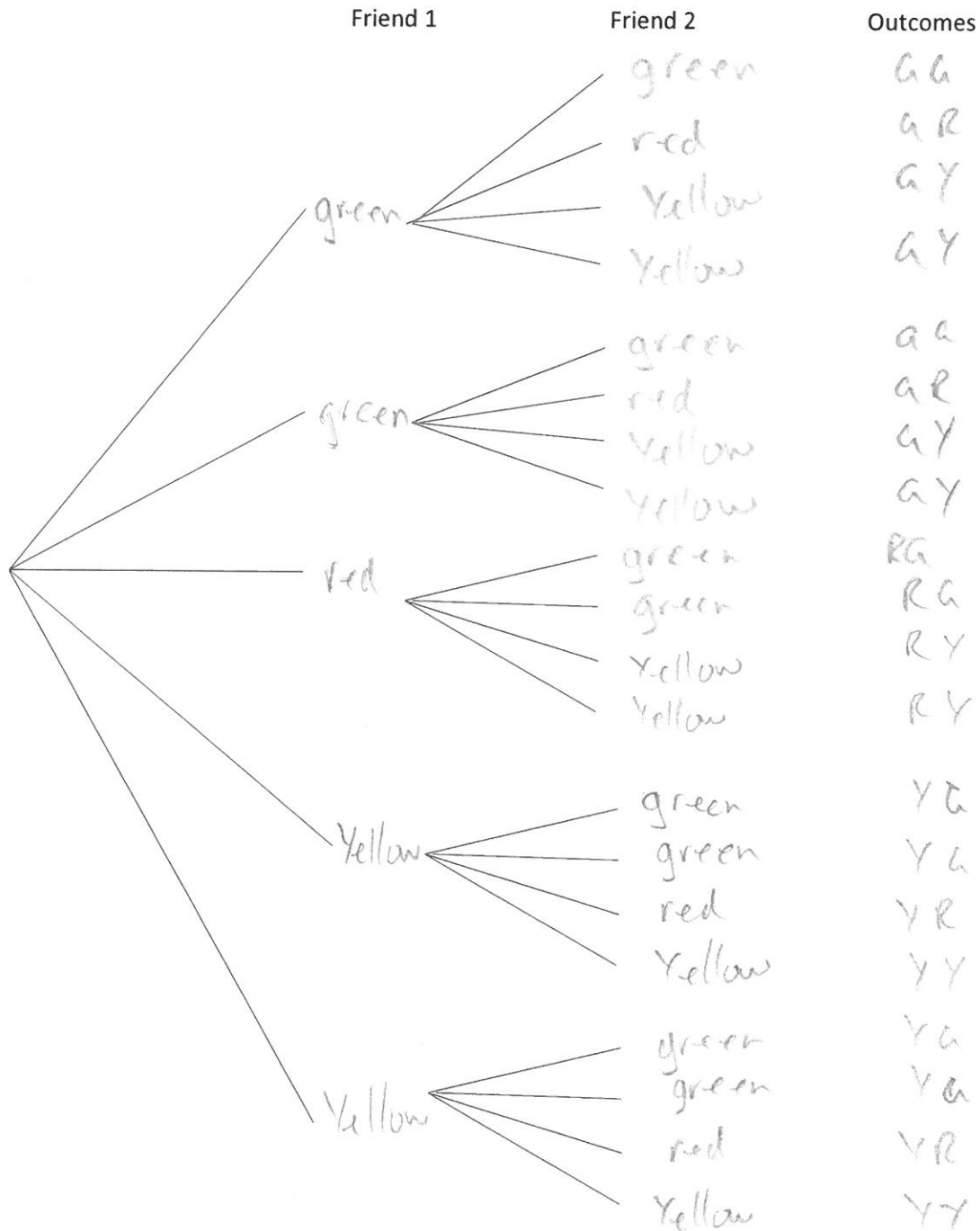
b) not having mechanical problems in the first year [1 mark]

$$rf = \frac{25\,000 - 750}{25\,000} = 0.97 \text{ or } 97\% \quad \checkmark$$

$$\text{or } 1 - 0.03 = 0.97$$

14. A friend has 5 snake lollies left in a bag, 2 green, 1 red and 2 yellow ones. He offers one snake each to his 2 friends.

a) Draw a tree diagram and list the possible outcomes of the possible choices for the 2 friends. [5 marks]



- $\frac{1}{2}$ for errors

If the first friend actually takes and eats 2 snakes, 1 green and 1 yellow, what are the chances of the second friend getting the red snake? [2 mark]

remaining is 1x green, 1x red, 1x yellow.

End of Test $Prob(\text{red}) = \frac{1}{3}$ ✓

